Craftax runs compared: achievements and what unlocks them

The plain DreamerV3 runs on main: 1.1M env steps each, size50m, scored on the same 30 fixed worlds with tools/death_eval.py. Family runs are indexed in README.md and documented on their branches. Updated 2026-11-08.

Every number is the mean episode-final achievement count over 30 evaluation episodes, ± its standard error over those worlds; unlock rates are the share of episodes in which an achievement fired; return is the mean raw Craftax return as a % of the maximum (226). Every checkpoint is scored on the same fixed worlds (env.reseed, audit B05), so comparisons are paired world by world. The error bars cover the worlds only, not training seeds.

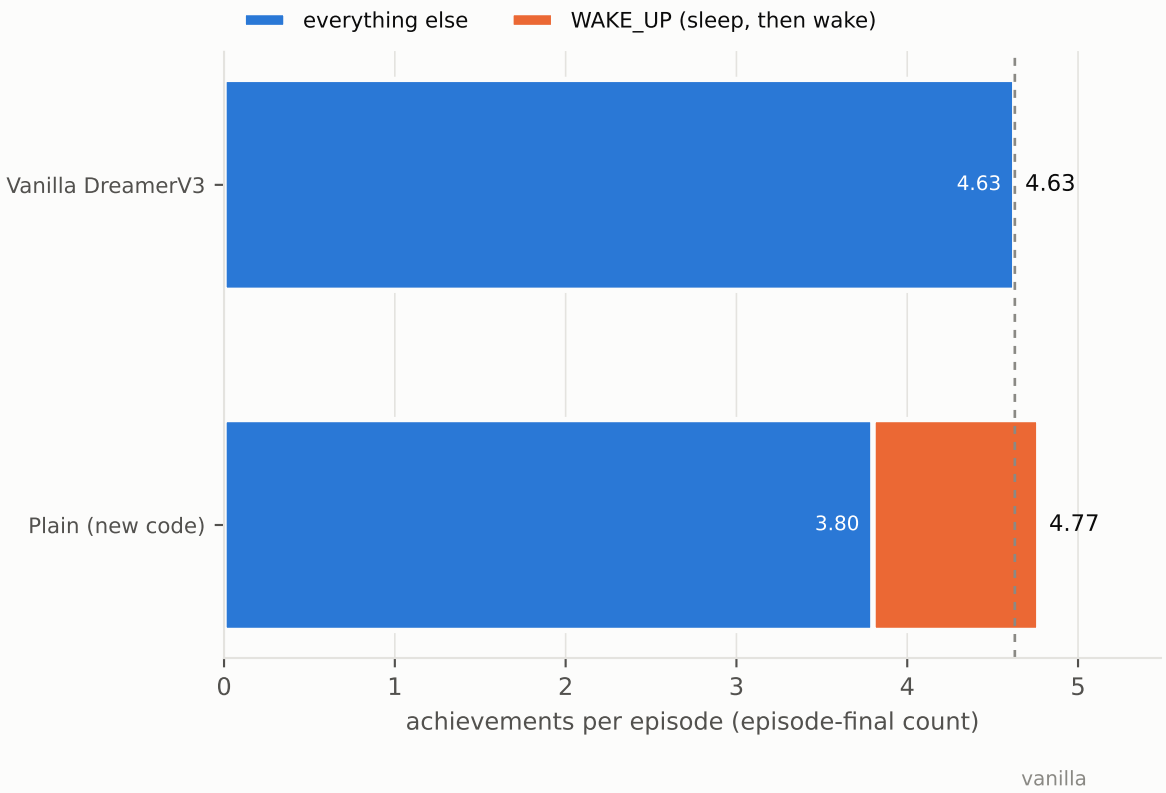
Plain DreamerV3 on the fixed code (abl_plain, seed 0) scores 4.77 ± 0.28, level with vanilla DreamerV3 on the original code (4.63 ± 0.22). Normalized return: 1.71% and 1.65%.

run	ach. ± SE	w/o wake	life	return	table	pickaxe	sword	stone	drink
Vanilla DreamerV3	4.63 ± 0.22	4.63	295	1.65%	63%	23%	3%	0%	77%
Plain (new code)	4.77 ± 0.28	3.80	265	1.71%	50%	3%	10%	0%	73%

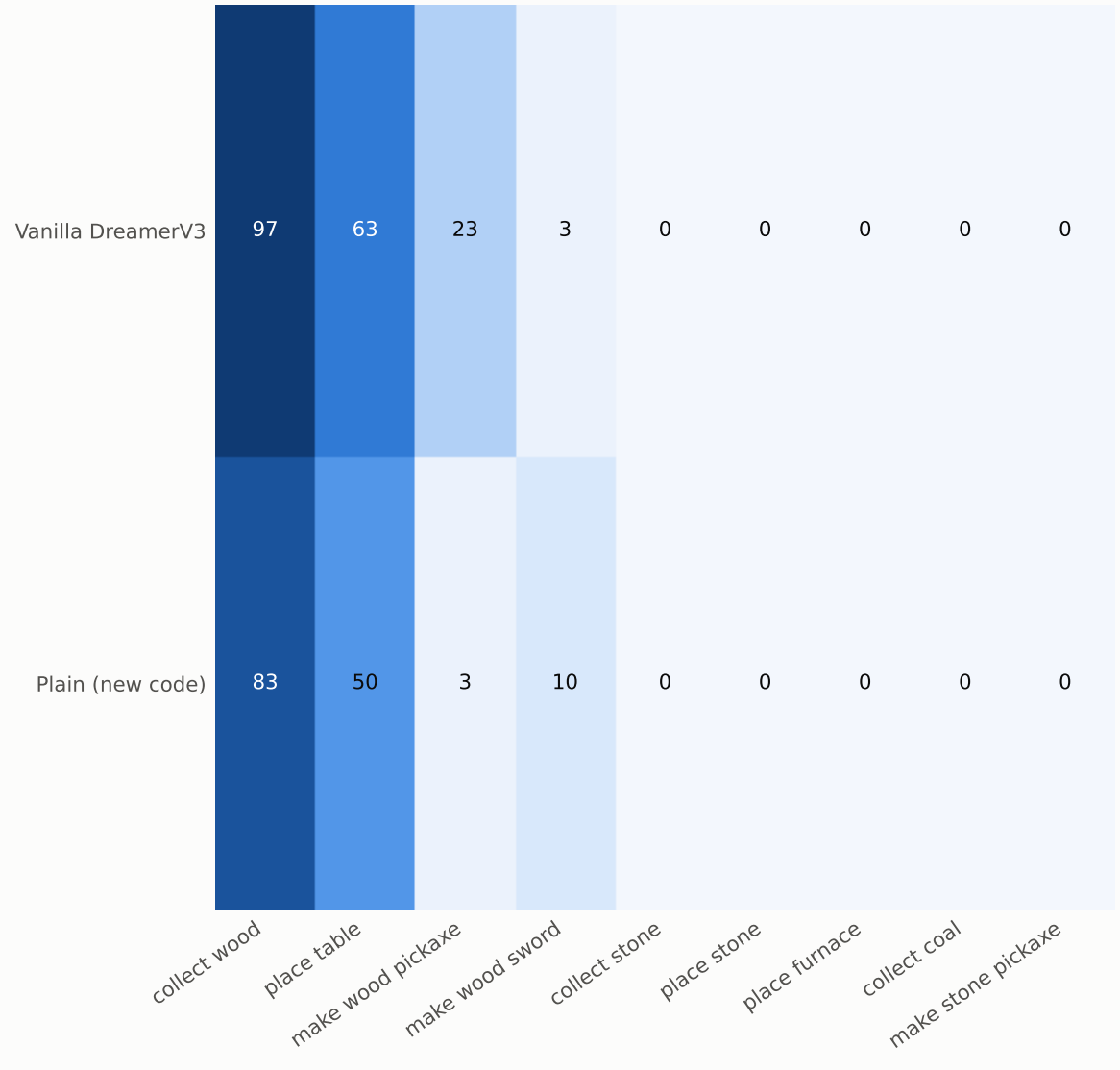
ach. ± SE = mean episode-final achievements ± standard error over the 30 worlds. w/o wake = the same minus WAKE_UP. life = mean lifespan in steps. return = raw Craftax return, % of 226. Rates are % of episodes. Vanilla DreamerV3: the original checkpoint (code before the 2026-10 fixes), re-scored on the fixed worlds. Plain (new code): this branch's code, `--configs craftax size50m --env.craftax.survival none`; the row shows seed 0.

Achievements per run, and the early tech tree

Mean episode-final achievements per run, split into WAKE_UP and the rest; unlock rate of the early tech-tree rungs

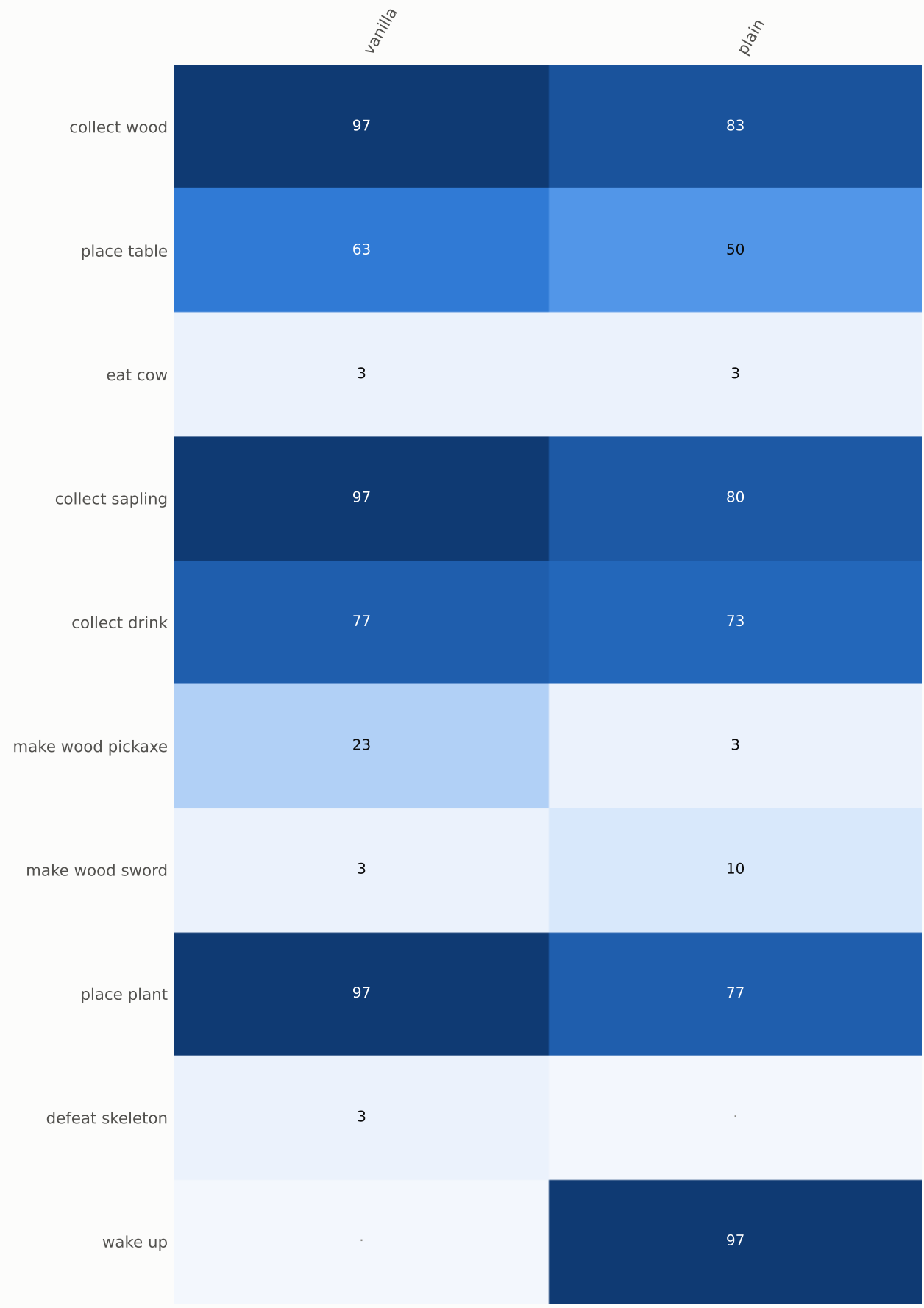


Unlock rate (% of episodes), early tech tree in dependency order



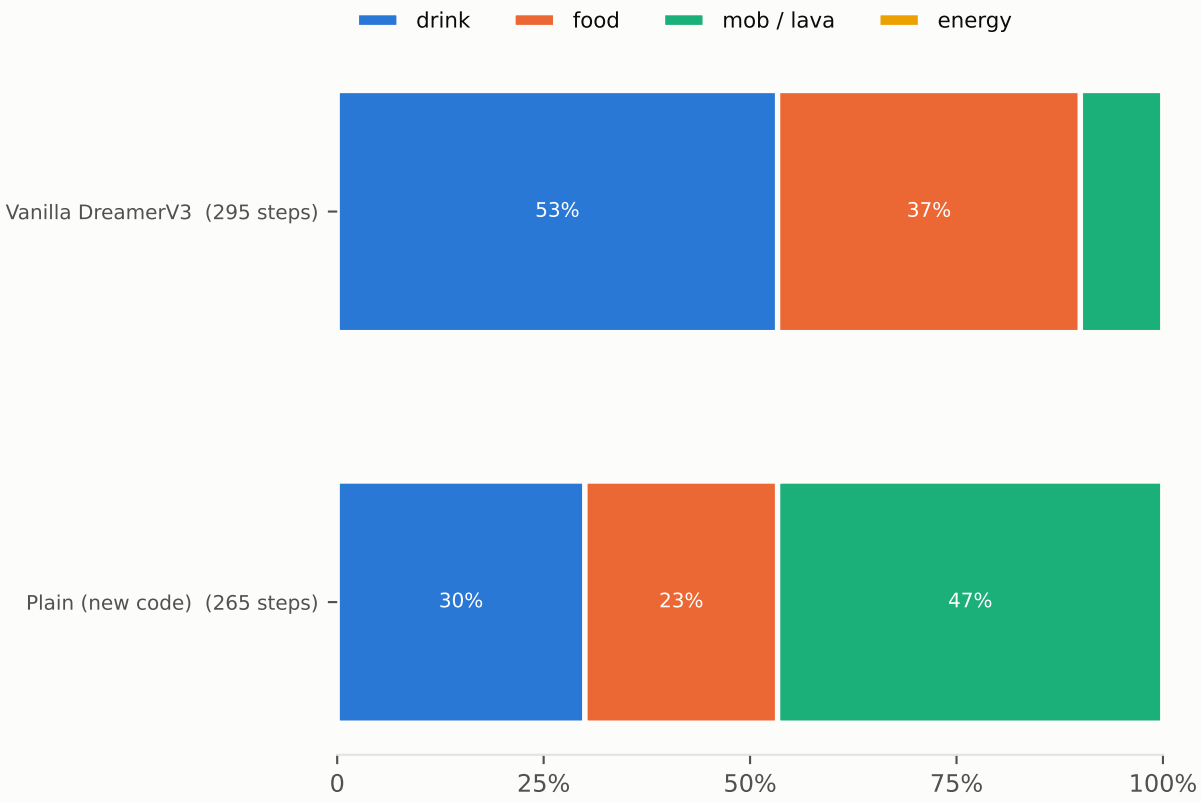
Every achievement any run unlocked

% of evaluation episodes in which each achievement fired. The other 57 of 67 were never unlocked by any run.



Cause of death and lifespan

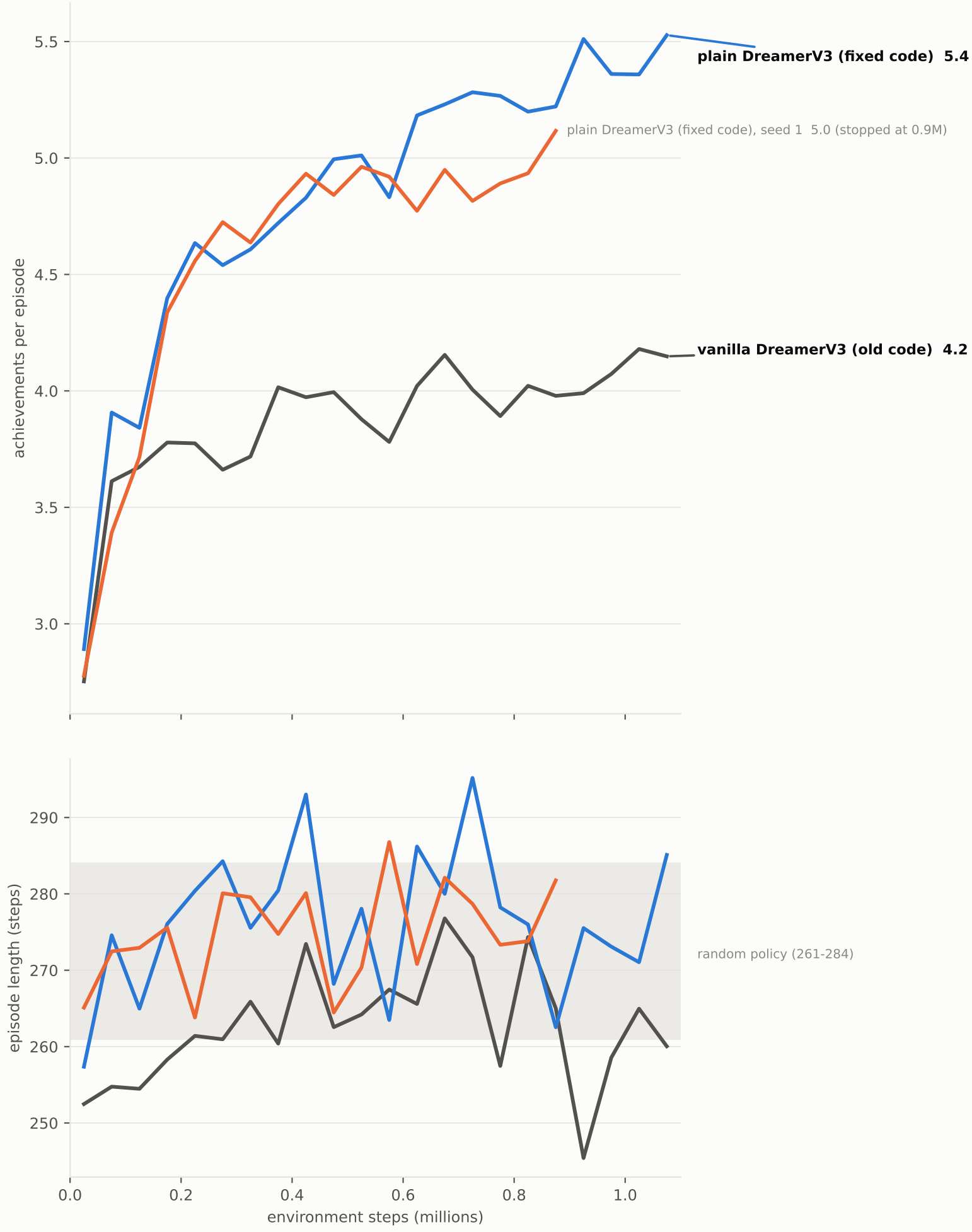
Cause of death (share of evaluation episodes) and lifespan; and, if the results list one, how much a run moves between its own checkpoints



Random policy lives 261-284 steps; doing nothing lives 333.

Training curves: every run

Rebuilt from each run's replay; 50k-step bins; value = mean of the last 100k steps.



What the results say, and what would settle the rest

The plain baseline

The fixed code does not move plain DreamerV3: 4.77 against 4.63, well inside the noise of a 30-world score (about ±0.55 between world sets; a gap under ~1.1 between single runs is not evidence). Both stay at the wood tier: no stone is collected in any episode, and lifespans (265-295 steps) sit near the random-policy band (261-284).

Where the other results are

Every agent built on top of this baseline (map model, goal manager, named goals, taste, rule mask) lives on its own family branch with its own results; README.md indexes them with their best scores. The full pre-split results, all runs side by side, are in the tag archive/full-2026-11-08.